# Seyit Kubilay ULUÇAY

Electrical & Electronics Engineer

Istanbul, Turkey | kubiulucay@gmail.com | +90 555 140 32 48
ulucay.me | linkedin.com/in/kubilayulucay | github.com/KubilayUlucay

## Education

**B.Sc. in Electrical and Electronics Engineering** 2021 – Jun 2025
Özyeğin University, Istanbul, Turkey
GPA: 2.97/4.00.
**Relevant coursework:** Electronics, mechatronics, industrial automation, power systems and renewable energy.

## Experience

**Quality Assurance Engineer, Accenture** Jan 2026 – Present
Industry X | Istanbul, Turkey

- Develop requirements-based Python/Gherkin black-box tests for clinical-laboratory automation at a global pharmaceutical company.
- Refactor test functions and state-change checks; deliver changes through reviewed GitLab merge requests.
- Run automated test suites locally and in existing GitLab/AWS pipelines; investigate test and suspected product issues.
- Maintain requirement, test-record and defect traceability in Codebeamer; submitted 40 log-backed issue reports over five months.

**R&D Intern, Beko** Jul – Aug 2024
Advanced Sensor Technologies, R&D | Istanbul, Turkey

- Built a motorized test rig and adapted acquisition code to collect sensor measurements across various environments and configurations.
- Preprocessed approximately 650,000 logged data points in Python/Excel using filtering and summary statistics; developed regression models in Minitab.
- Integrated the model into embedded firmware and achieved 83.5% floor-type prediction success in project trials.

**Student Services (Part-Time), Özyeğin University** Jun – Sep 2023
Student Services | Istanbul, Turkey

- Developed a Power Automate workflow for course assignments, reducing manual processing time by approximately 50%.

**Undergraduate Assistant, Özyeğin University** Mar – Jun 2023
Faculty of Engineering | Istanbul, Turkey

- Supported NVIDIA Jetson Nano-based vehicle assembly and hardware troubleshooting for an autonomous-driving course.

## Engineering Projects

**Battery Health and Performance Management System** 2024 – 2025
STM32F407, MATLAB/Simulink, instrumentation | Senior-year team project

- Developed and fabricated a four-cell lithium-ion sensing and capacitor-based active-balancing system.
- Implemented STM32 monitoring and balancing logic, integrating INA333 signal conditioning and TMP36GZ temperature sensing.
- Built all MATLAB/Simulink charge-transfer models; the reported charging simulation reduced summed cell-voltage imbalance from 302.94 mV to 2.93 mV (99.0% reduction) at approximately 27,040 s of simulated time.
- Integrated the sensing and switching circuitry with the controller.

**Remote-Controlled Axial-Flux Motor** 2024
Arduino Nano, NRF24L01, ESC, 3D-printed mechanics | Two-person build

- Implemented transmitter/receiver electronics and Arduino firmware to send potentiometer commands through paired NRF24L01 modules to a motor-side ESC.
- Shared winding and assembly of a nine-coil motor, completed and bench-demonstrated in two months.

**Servo-Actuated Three-Axis Camera Gimbal** 2024
Arduino Nano, MPU6050, servo control, 3D printing | Two-person build

- Implemented Arduino servo actuation and MPU6050 readout, and shared assembly and integration.
- Cross-checked phone and onboard gyroscope readings during the build.

**Boundary-Scanning Zumo Robot** 2024
Arduino C/C++, IR and reflectance sensors | Team project

- Implemented reflectance-based boundary navigation, IR detection counting and repeated-scan logic.
- Reported the largest observed detection count from repeated scans through LED and buzzer feedback.

**PLC Sequences and Operator Interfaces** 2024
Schneider EcoStruxure Control Expert, PLC, HMI | Team simulation project

- Developed state-based traffic-light and timed claw-game sequences with operator HMI controls.
- Checked operator inputs, mode transitions, timers and internal counters in Control Expert simulation.

**Power Flow and Economic Dispatch Application** 2022
MATLAB, App Designer | Individual academic project

- Built a four-bus network interface with generator settings, demand inputs and branch-flow outputs.
- Solved three-generator dispatch from equal marginal costs and demand balance, then calculated quadratic costs.

**Off-Grid Residential Photovoltaic Design** 2025
PVsyst, solar sizing, energy modeling | Team design study

- Sized a 4.64 kW array and 48 V storage system around household demand, Kocaeli weather and three days of autonomy.
- Simulated 3,043.3 kWh annual supply against 3,044.1 kWh demand in PVsyst, covering 99.97% of the modeled load.
- Estimated 389,000 TL capital cost using May 2025 prices; found grid electricity cheaper where available.

**Wind-Farm Interconnection Analysis** 2022
PowerWorld Simulator | Individual academic study

- Screened 27 connections for a hypothetical 200 MW wind farm; only two met the study's feasibility criteria.
- Recommended Experiment 9: CONDOR links to PETE and PAI, with about 91% link loading and 20.11 MW total modeled system losses, balancing loading feasibility and estimated five-year cost.

## Technical Skills and Laboratory Practice

**Programming:** Python, MATLAB, Arduino and STM32 C/C++.
**Tools:** MATLAB/Simulink, KiCad, Control Expert, PVsyst, PowerWorld, Minitab, Power Automate, Codebeamer.
**Version control and testing:** GitLab, merge requests, Gherkin, black-box testing, requirements traceability.
**Hardware and bench work:** STM32, Arduino, NRF24L01, I2C, oscilloscope, multimeter, DC supply, breadboard measurements.
**Mechatronics studies:** Analog/reflectance sensing, oscillator timing, two-board I2C communication and observed proportional motor-position responses.

## Leadership, Awards and Languages

**IEEE Özyeğin University Student Branch** 2021 – 2025
Chairman, 2023–2024; President of the Supervisory Board, 2024–2025.
During branch leadership, annual events grew from 4 to 23 and membership from 103 to 652 over two years.

**Robotics competitions:** Team captain; Mini Sumo 1st place at KELEBEKRO 2019; Micro Sumo 2nd and 3rd places at ITURO 2019.

**Languages:** Turkish (native), English (fluent), German (beginner).